

Machine Learning Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

1. Introduction to Machine Learning

- What is Machine Learning?
- Artificial Intelligence vs Machine Learning
- Machine Learning vs Deep Learning
- Traditional Programming vs Machine Learning
- Applications and Real-World Use Cases
- Dataset, Observation, Sample, Features and Target
- Labels, Training, Prediction and Inference
- Models, Parameters and Hyperparameters
- Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning
- Machine Learning Workflow

2. Data Preparation for Machine Learning

- Data Quality
- Missing Values
- Duplicate Records
- Incorrect and Inconsistent Values
- Mean, Median and Mode Imputation
- KNN Imputation
- Iterative Imputation
- Outlier Detection using IQR, Z-Score and Percentiles
- Isolation Forest for Outlier Detection
- Outlier Treatment: Removal, Capping, Winsorization and Transformation

3. Data Encoding

- Nominal, Ordinal, Binary and High-Cardinality Variables
- Label Encoding
- Ordinal Encoding
- One-Hot Encoding
- Dummy Encoding
- Binary Encoding
- Frequency and Count Encoding
- Target Encoding
- Unseen Categories
- Rare Categories
- Data Leakage Considerations

4. Feature Scaling and Transformation

- Why Feature Scaling is Required
- Standardization
- Min-Max Scaling
- MaxAbs Scaling
- Robust Scaling
- Log Transformation
- Square Root Transformation
- Reciprocal Transformation

- Box-Cox
- Yeo-Johnson
- Quantile Transformation

5. Exploratory Analysis for Machine Learning

- Dataset Understanding
- Feature Types
- Target Variable Analysis
- Class Distribution
- Correlation and Covariance
- Scatter Plots and Pairwise Relationships
- Multicollinearity
- Variance Inflation Factor

6. Feature Engineering

- Feature Engineering Fundamentals
- Numerical Feature Creation
- Ratios and Differences
- Aggregations
- Polynomial Features
- Interaction Features
- Categorical Feature Engineering
- Rare Category Handling
- Date-Time Features
- Binning and Discretization

7. Feature Selection

- Importance of Feature Selection
- Filter Methods
- Correlation
- Variance Threshold
- Chi-Square Test
- ANOVA F-Test
- Mutual Information
- Wrapper Methods
- Forward Selection
- Backward Elimination
- Recursive Feature Elimination
- RFECV
- Embedded Methods
- Lasso
- Tree-Based Feature Importance
- Permutation Importance

8. Dataset Splitting

- Train-Test Split
- Train-Validation-Test Split
- Stratified Splitting
- Time-Based Splitting
- Random State

9. Bias, Variance and Generalization

- Generalization
- Underfitting
- Overfitting
- High Bias
- High Variance
- Bias-Variance Trade-Off
- Reducing Overfitting

10. Supervised Learning

- Supervised Learning Fundamentals
- Regression
- Classification
- Binary Classification
- Multiclass Classification
- Multilabel Classification

11. Linear Regression

- Simple Linear Regression
- Multiple Linear Regression
- Slope and Intercept
- Residuals
- Ordinary Least Squares
- Regression Assumptions
- Residual Analysis

12. Polynomial Regression

- Nonlinear Relationships
- Polynomial Features
- Degree Selection
- Underfitting and Overfitting

13. Regularized Regression

- Regularization Fundamentals
- Ridge Regression
- Lasso Regression
- Elastic Net
- L1 and L2 Penalties
- RidgeCV, LassoCV and ElasticNetCV

14. Regression Model Evaluation

- Mean Absolute Error
- Mean Squared Error
- Root Mean Squared Error
- R-Squared
- Adjusted R-Squared
- Mean Absolute Percentage Error
- Metric Selection by Business Context

15. Logistic Regression

- Binary Classification
- Sigmoid Function
- Probability and Log Odds
- Decision Boundary
- Threshold Adjustment
- Coefficients and Regularization
- Multiclass Logistic Regression
- One-vs-Rest
- Multinomial Classification

16. K-Nearest Neighbors

- Instance-Based and Lazy Learning
- Euclidean, Manhattan and Minkowski Distance
- Choosing K
- KNN Classification
- KNN Regression
- Weighted Neighbors
- Scaling Sensitivity
- Curse of Dimensionality

17. Naive Bayes

- Bayes' Theorem
- Prior, Likelihood and Posterior
- Feature Independence Assumption
- Gaussian Naive Bayes
- Multinomial Naive Bayes
- Bernoulli Naive Bayes

18. Decision Trees

- Root, Decision and Leaf Nodes
- Gini Impurity
- Entropy
- Information Gain
- Classification Trees
- Regression Trees
- max_depth
- min_samples_split
- min_samples_leaf
- max_features
- Pre-Pruning and Post-Pruning
- Cost Complexity Pruning

19. Support Vector Machines

- Hyperplane
- Support Vectors
- Margin
- Maximum Margin Classifier
- Soft Margin
- C Parameter

- Kernel Trick
- Linear, Polynomial, RBF and Sigmoid Kernels
- Gamma
- Support Vector Regression

20. Classification Model Evaluation

- Confusion Matrix
- True Positive, True Negative, False Positive and False Negative
- Accuracy
- Precision
- Recall
- Specificity
- F1 Score
- ROC Curve
- AUC
- Precision-Recall Curve
- Log Loss
- Threshold Selection

21. Ensemble Learning

- Ensemble Fundamentals
- Weak and Strong Learners
- Bagging
- Boosting
- Voting
- Stacking

22. Bagging

- Bootstrap Sampling
- Multiple Base Models
- Bagging Classifier
- Bagging Regressor
- Out-of-Bag Evaluation

23. Random Forest

- Bootstrap Samples
- Random Feature Selection
- Random Forest Classification
- Random Forest Regression
- Hyperparameters
- Feature Importance
- Permutation Importance
- OOB Score

24. Boosting

- Sequential Learning
- Correcting Previous Errors
- AdaBoost
- Gradient Boosting
- Residual Learning

- Learning Rate
- Number of Estimators
- Gradient Boosting Regression and Classification

25. XGBoost

- Gradient Boosted Trees
- Regularized Boosting
- Gradient Optimization
- n_estimators
- learning_rate
- max_depth
- min_child_weight
- subsample
- colsample_bytree
- gamma
- reg_alpha
- reg_lambda
- Classification and Regression
- Feature Importance
- Early Stopping

26. LightGBM

- Leaf-Wise Growth
- Histogram-Based Learning
- Feature Binning
- num_leaves
- max_depth
- learning_rate
- n_estimators
- min_child_samples
- feature_fraction
- bagging_fraction
- Classification and Regression

27. CatBoost

- Categorical Feature Handling
- Ordered Boosting
- Target Statistics
- Classification and Regression
- iterations
- depth
- learning_rate
- l2_leaf_reg

28. Voting and Stacking

- Hard Voting
- Soft Voting
- Probability Averaging
- Voting Regressor
- Base Learners
- Meta Learner

- Out-of-Fold Predictions
- Blending

29. Imbalanced Machine Learning

- Majority and Minority Classes
- Limitations of Accuracy
- Random Oversampling
- Random Undersampling
- SMOTE
- Borderline-SMOTE
- ADASYN
- Tomek Links
- Class Weights
- Cost-Sensitive Learning
- Precision, Recall, F1, ROC-AUC and PR-AUC

30. Unsupervised Learning

- Unlabeled Data
- Pattern Discovery
- Clustering
- Dimensionality Reduction
- Anomaly Detection

31. K-Means Clustering

- Clusters and Centroids
- Distance
- Centroid Initialization
- Assignment and Update
- Convergence
- Elbow Method
- Silhouette Score
- K-Means++
- Limitations

32. Hierarchical Clustering

- Agglomerative Clustering
- Divisive Clustering
- Euclidean and Manhattan Distance
- Single, Complete, Average and Ward Linkage
- Dendrogram

33. DBSCAN

- Density-Based Clustering
- Epsilon
- Minimum Samples
- Core Points
- Border Points
- Noise Points
- Arbitrary-Shaped Clusters

34. Gaussian Mixture Models

- Probabilistic Clustering
- Gaussian Components
- Expectation-Maximization
- Hard vs Soft Clustering
- AIC
- BIC

35. Clustering Evaluation

- Inertia
- Silhouette Score
- Davies-Bouldin Index
- Calinski-Harabasz Score
- Adjusted Rand Index
- Normalized Mutual Information
- Cluster Profiling and Interpretation

36. Dimensionality Reduction

- Curse of Dimensionality
- Principal Component Analysis
- Covariance Matrix
- Eigenvalues and Eigenvectors
- Explained Variance
- Component Selection
- t-SNE
- UMAP

37. Anomaly Detection

- Point, Contextual and Collective Anomalies
- Z-Score
- IQR
- Isolation Forest
- Local Outlier Factor
- One-Class SVM
- Fraud, Network and Manufacturing Applications

38. Cross-Validation

- Why Cross-Validation?
- K-Fold
- Stratified K-Fold
- Leave-One-Out
- Repeated K-Fold
- Time-Series Cross-Validation
- Expanding and Rolling Windows

39. Hyperparameter Optimization

- Parameters vs Hyperparameters
- Manual Tuning
- GridSearchCV

- RandomizedSearchCV
- Bayesian Optimization
- Optuna
- Study, Trial, Search Space and Pruning
- Choosing the Best Model

40. Machine Learning Pipelines

- Data Preprocessing Pipeline
- Feature Engineering Pipeline
- Modeling Pipeline
- Scikit-Learn Pipeline
- ColumnTransformer
- Custom Transformers
- Pipeline with Cross-Validation
- Pipeline with Hyperparameter Tuning

41. Data Leakage

- Training-Test Contamination
- Target Leakage
- Preprocessing Leakage
- Scaling Before Splitting
- Imputation Before Splitting
- Feature Selection Before Splitting
- Temporal Leakage
- Preventing Leakage with Pipelines

42. Model Calibration and Threshold Optimization

- Predicted Probabilities
- Calibration
- Reliability Curves
- Platt Scaling
- Isotonic Regression
- Default vs Custom Threshold
- Precision-Recall Trade-Off
- Business Cost-Based Threshold Selection

43. Model Interpretability and Explainability

- Transparent vs Black-Box Models
- Global and Local Explainability
- Model Coefficients
- Decision Tree Interpretation
- Feature Importance
- Permutation Importance
- Partial Dependence Plots
- ICE
- LIME
- SHAP
- Summary, Dependence and Waterfall Plots

44. Error Analysis

- Model Error Analysis
- False Positives and False Negatives
- Regression Residual Analysis
- Large Error Investigation
- Segment-Based Analysis
- Improving Features, Data and Algorithms

45. Cost-Sensitive Machine Learning

- False Positive Cost
- False Negative Cost
- Cost Matrix
- Class Weighting
- Threshold Optimization
- Fraud, Credit Risk, Churn and Screening Applications

46. Semi-Supervised Learning

- Labeled and Unlabeled Data
- Self-Training
- Pseudo Labels
- Confidence Thresholds
- Label Propagation
- Label Spreading

47. Reinforcement Learning Fundamentals

- Agent, Environment, State, Action and Reward
- Policy
- Immediate and Cumulative Reward
- Discount Factor
- Exploration vs Exploitation
- Markov Decision Process
- Q-Learning
- Q-Table
- Bellman Equation
- Epsilon-Greedy Strategy

48. Recommendation Systems

- Users, Items and Ratings
- User-Item Matrix
- Popularity-Based Recommendations
- Content-Based Filtering
- Cosine Similarity
- User-Based Collaborative Filtering
- Item-Based Collaborative Filtering
- Matrix Factorization
- Singular Value Decomposition
- Precision@K, Recall@K, MAP and RMSE

49. Machine Learning Project Workflow

1. Problem Definition
2. Dataset Understanding

3. Target Identification
4. Data Cleaning
5. Exploratory Analysis
6. Train-Test Splitting
7. Feature Engineering
8. Encoding
9. Scaling
10. Baseline Model
11. Multiple Model Training
12. Cross-Validation
13. Hyperparameter Optimization
14. Model Evaluation
15. Error Analysis
16. Feature Importance
17. Explainability
18. Final Model Selection
19. Prediction
20. Documentation

50. Mini Projects

- House Price Prediction
- Customer Churn Prediction
- Loan Approval Prediction
- Credit Risk Prediction
- Customer Segmentation
- Fraud Detection
- Employee Attrition Prediction
- Product Recommendation System
- Disease Risk Classification

51. Capstone Project

- Problem Definition
- Data Preparation
- Exploratory Analysis
- Feature Engineering
- Baseline Modeling
- Multiple Algorithms
- Cross-Validation
- Hyperparameter Tuning
- Threshold Optimization
- Model Evaluation
- Error Analysis
- Explainability
- Final Model

- Documentation

52. Interview Preparation

- ML Fundamentals
- Regression
- Classification
- Ensemble Learning
- Unsupervised Learning
- Model Evaluation
- Hyperparameter Optimization
- Feature Engineering and Selection
- Imbalanced Data
- Data Leakage
- Explainability
- ML Coding Exercises
- Dataset Case Studies
- Mock Interviews
- Python
- NumPy
- Pandas
- SciPy
- Scikit-Learn
- imbalanced-learn
- XGBoost
- LightGBM
- CatBoost
- Optuna
- Matplotlib
- Seaborn
- SHAP
- LIME
- MLflow
- Joblib
- FastAPI
- Streamlit
- Docker
- Git
- GitHub
- Jupyter Notebook
- Google Colab
- Azure / AWS / GCP for model deployment

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